Synopsis

Air Quality Level Classification System

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- **Problem Description:**

  - Air pollution has become a major environmental issue affecting human health, climate, and ecosystems. Rapid industrialization, urbanization, and increasing vehicle emissions have significantly increased pollution levels in many regions.
  - Poor air quality can cause serious health problems such as asthma, respiratory diseases, lung infections, and other long-term health issues. It also harms plants, animals, and disturbs the ecological balance.
  - To monitor pollution levels, environmental monitoring systems collect data from sensors measuring parameters such as Carbon Monoxide (CO), Nitrogen Dioxide (NO_2), temperature, and humidity**.**
  - However, these systems generate large volumes of data continuously, making manual analysis difficult, time-consuming, and inefficient.
  - Traditional monitoring methods often struggle to provide quick and accurate classification of air quality levels, which can delay important decisions related to pollution control.
  - To solve this problem, machine learning techniques can be used to automatically analyze environmental data and identify patterns in pollution levels.
  - Using algorithms such as K-Nearest Neighbors (KNN) and Logistic Regression, the system can classify air quality into categories like Good, Moderate, Poor, or Severe.
  - This automated system can support real-time pollution monitoring, smart city systems, and better environmental decision-making.

- **Steps of Implementation:**

Data Collection- Load air quality dataset containing pollutants like CO, NO_2, PM2.5 and other environmental attributes.
Data Preprocessing- Handle missing values, remove unnecessary columns, and clean noisy data.
Feature Selection- Select important pollution indicators that contribute most to AQI prediction.
Data Scaling- Normalize numeric values using scaling techniques to improve model performance.
Model Training- Train classification models such as KNN and Logistic Regression.
Model Testing- Test the trained model on unseen data to evaluate prediction performance.
Evaluation- Use evaluation metrics like accuracy and precision to measure model effectiveness.

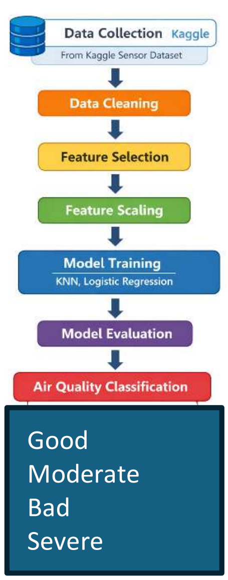

Fig1. Architecture

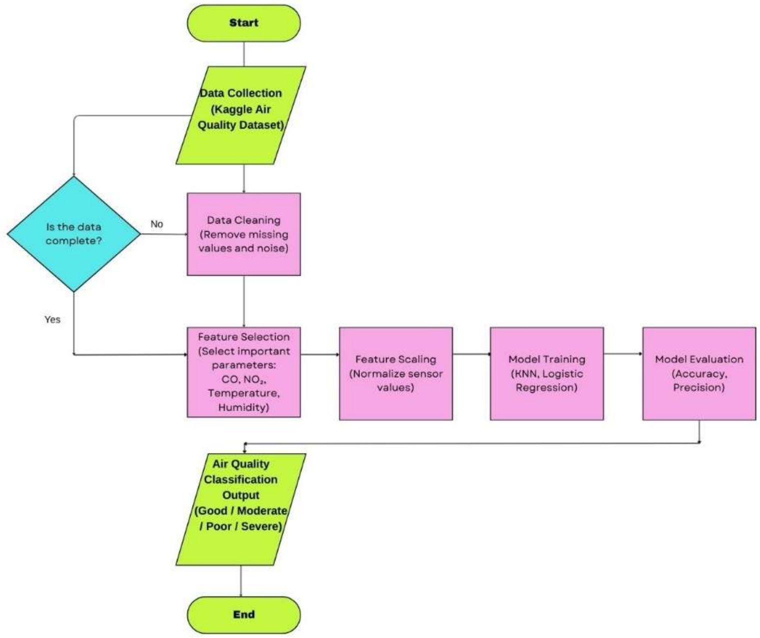

Fig.2 Workflow Diagram Flowchart

- **Expected Modules/ Libraries to be used :**

NumPy:
A numerical computing library used to perform mathematical operations on arrays and matrices efficiently.
Pandas:
A powerful data manipulation library used for reading datasets, cleaning data and handling tabular data structures like DataFrames.
Matplotlib:
A visualization library used to create graphs, charts and plots for understanding patterns in air quality data.
Seaborn:
A statistical visualization library built on top of Matplotlib which helps in creating more informative plots.
Scikit-learn:
A machine learning library that provides algorithms such as KNN and Logistic Regression along with tools for preprocessing and evaluation.

- **Expected Evaluation metrics and Plots to be used:**

Accuracy- Measures the percentage of correct predictions made by the model.

Precision- Evaluates how many predicted positive observations are actually correct.

Confusion Matrix- Helps visualize classification performance by showing correct and incorrect predictions.

Visualization Plots- Correlation heatmaps, feature distribution plots, and model comparison graphs will be used to understand the dataset and model performance.

- **Required topics from the subject:**

Strings- Used for handling categorical labels such as AQI categories.
Files- Reading and writing dataset files such as CSV format.

Data Structures / Arrays- Used for storing and processing numerical sensor data.

- **Platform Required:**

Python Jupyter Notebook, Google Colab, Visual Studio Code for implementing and testing machine learning models.

- **Books and Link Sources:**

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow- Aurélien Géron.
Introduction to Machine Learning with Python- Andreas C. Müller.
https://scikit-learn.org
https://pandas.pydata.org
https://www.kaggle.com (Air Quality datasets)